

Comparison of Various Models on our Dataset

1. Performance Comparison

Model	RMSE (Mean $\pm$ Std)	MAE (Mean $\pm$ Std)	Accuracy	Recall	F1-score
SVD	0.9784 $\pm$ 0.0852	0.7929 $\pm$ 0.0765	0.770	0.630	0.747
KNN Basic	1.0339 $\pm$ 0.0765	0.8146 $\pm$ 0.0289	0.730	0.611	0.710
NMF	0.6315 $\pm$ 0.0767	0.4641 $\pm$ 0.0521	0.830	0.704	0.817
SlopeOne	1.0274 $\pm$ 0.0703	0.8039 $\pm$ 0.0494	0.790	0.648	0.769
CoClustering	0.7016 $\pm$ 0.0450	0.5404 $\pm$ 0.0338	0.830	0.741	0.825
BaselineOnly	1.0505 $\pm$ 0.0414	0.8729 $\pm$ 0.0394	0.760	0.593	0.727
KNN Pearson Baseline	0.5976 $\pm$ 0.0099	0.4487 $\pm$ 0.0153	0.850	0.759	0.845

2. Cross-Validation Results Comparison

Model	Metric	Fold 1	Fold 2	Fold 3	Mean	Std
SVD	RMSE (testset)	0.9897	0.8689	1.0766	0.9784	0.0852
	MAE (testset)	0.7940	0.6987	0.8861	0.7929	0.0765
KNN Basic	RMSE (testset)	0.9908	1.1415	0.9695	1.0339	0.0765
	MAE (testset)	0.8138	0.8504	0.7796	0.8146	0.0289
NMF	RMSE (testset)	0.7366	0.5557	0.6021	0.6315	0.0767
	MAE (testset)	0.5354	0.4127	0.4440	0.4641	0.0521
SlopeOne	RMSE (testset)	1.0409	1.1060	0.9354	1.0274	0.0703
	MAE (testset)	0.8161	0.8574	0.7382	0.8039	0.0494
CoClustering	RMSE (testset)	0.7356	0.6380	0.7311	0.7016	0.0450
	MAE (testset)	0.5749	0.4945	0.5519	0.5404	0.0338

BaselineOnly	RMSE (testset)	1.0692	0.9932	1.0892	1.0505	0.0414
	MAE (testset)	0.9010	0.8172	0.9005	0.8729	0.0394
KNN Pearson Baseline	RMSE (testset)	0.6087	0.5846	0.5996	0.5976	0.0099
	MAE (testset)	0.4287	0.4516	0.4659	0.4487	0.0153

RAWDATA

Table 1: Cross-Validation Results of SVD Model

Metric	Fold 1	Fold 2	Fold 3	Mean	Std
RMSE (testset)	0.9897	0.8689	1.0766	0.9784	0.0852
MAE (testset)	0.7940	0.6987	0.8861	0.7929	0.0765
Fit time	0.00	0.00	0.00	0.00	0.00
Test time	0.00	0.00	0.00	0.00	0.00

Additional Evaluation Metrics

Accuracy: 0.770

Recall: 0.630

F1-score: 0.747

Table 2: Cross-Validation Results of KNN Basic Model

Metric	Fold 1	Fold 2	Fold 3	Mean	Std
RMSE (testset)	0.9908	1.1415	0.9695	1.0339	0.0765
MAE (testset)	0.8138	0.8504	0.7796	0.8146	0.0289
Fit time	0.00	0.00	0.00	0.00	0.00
Test time	0.00	0.00	0.00	0.00	0.00

Additional Evaluation Metrics

Accuracy: 0.730

Recall: 0.611

F1-score: 0.710

Table 3: Cross-Validation Results of NMF Model

Metric	Fold 1	Fold 2	Fold 3	Mean	Std
RMSE (testset)	0.7366	0.5557	0.6021	0.6315	0.0767
MAE (testset)	0.5354	0.4127	0.4440	0.4641	0.0521
Fit time	0.00	0.00	0.00	0.00	0.00
Test time	0.00	0.00	0.00	0.00	0.00

Additional Evaluation Metrics

Accuracy: 0.830

Recall: 0.704

F1-score: 0.817

Table 4: Cross-Validation Results of SlopeOne Model

Metric	Fold 1	Fold 2	Fold 3	Mean	Std
RMSE (testset)	1.0409	1.1060	0.9354	1.0274	0.0703
MAE (testset)	0.8161	0.8574	0.7382	0.8039	0.0494
Fit time	0.00	0.00	0.00	0.00	0.00
Test time	0.00	0.00	0.00	0.00	0.00

Additional Evaluation Metrics

Accuracy: 0.790

Recall: 0.648

F1-score: 0.769

Table 5: Cross-Validation Results of CoClustering Model

Metric	Fold 1	Fold 2	Fold 3	Mean	Std
RMSE (testset)	0.7356	0.6380	0.7311	0.7016	0.0450
MAE (testset)	0.5749	0.4945	0.5519	0.5404	0.0338
Fit time	0.00	0.00	0.00	0.00	0.00
Test time	0.00	0.00	0.00	0.00	0.00

Additional Evaluation Metrics

Accuracy: 0.830

Recall: 0.741

F1-score: 0.825

Table 6: Cross-Validation Results of BaselineOnly Model

Metric	Fold 1	Fold 2	Fold 3	Mean	Std
RMSE (testset)	1.0692	0.9932	1.0892	1.0505	0.0414
MAE (testset)	0.9010	0.8172	0.9005	0.8729	0.0394
Fit time	0.00	0.00	0.00	0.00	0.00
Test time	0.00	0.00	0.00	0.00	0.00

Additional Evaluation Metrics

Accuracy: 0.760

Recall: 0.593

F1-score: 0.727

Table 7: Cross-Validation Results of KNN – PearsonBaseline Model

Metric	Fold 1	Fold 2	Fold 3	Mean	Std
RMSE (testset)	0.6087	0.5846	0.5996	0.5976	0.0099
MAE (testset)	0.4287	0.4516	0.4659	0.4487	0.0153
Fit time	0.00	0.00	0.00	0.00	0.00
Test time	0.00	0.00	0.00	0.00	0.00

Additional Evaluation Metrics

Accuracy: 0.850
Recall: 0.759
F1-score: 0.845